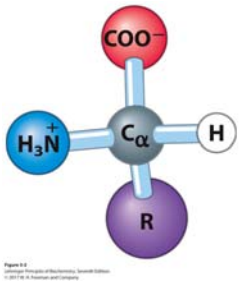
Amino acids, peptides and proteins

- Proteins are involved in all cellular processes, performing an extensive variety of functions. They are the *executors* of genomic information.
- They are found in all parts of the cell: plasma membrane, cytoplasm, organelles, and nucleus.
- Proteins or peptides are polymers of amino acids (AAs). All living organisms, from bacteria to humans, synthesize proteins from a repertoire of 20 standard amino acids.
- The key to understanding how thousands of different proteins—each with unique functions—can be built from only 20 amino acids lies in understanding the physicochemical properties of each AA.

Chemical structure of Aa

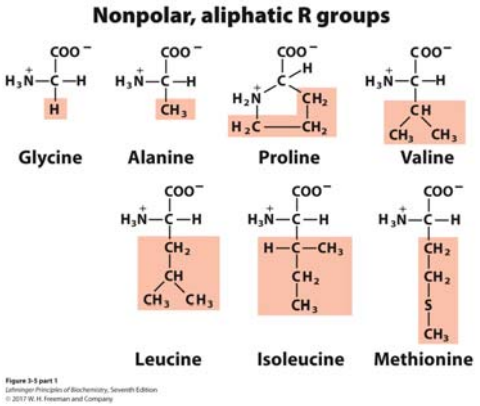
- In the structure of the 20 Aa we have to distinguish the elements common to all of them and the differential elements.
- Common to all AAs is the presence of a carboxyl group and an amino group, both attached to the same carbon, which we call the α carbon (Cα)



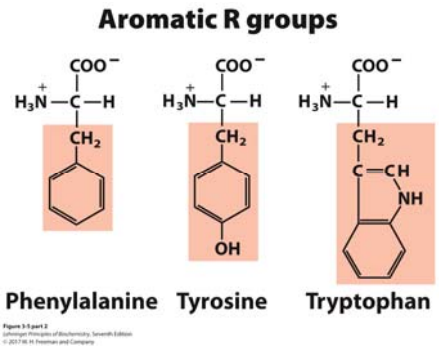
- The differential elements between them are the structure, size and physicochemical properties of the chemical structure attached to the Cα such as the electrical charge, hydrophobicity and hydrophilicity. This element is called the R group or side chain
- When analyzing Cα in detail, we realize that it is a chiral carbon that can have two configurations, that is, two stereoisomers
- Both stereoisomers are mirror images (D or L depending on the position of the NH_3^+ group). In this case, both stereoisomers are called enantiomers.
- In nature, all proteins (with very rare exceptions) contain Aa that belong to the L enantiomer
- Knowledge of the chemical properties of R groups is essential to understand the functions that proteins are capable of performing. Aa are classified based on these properties.

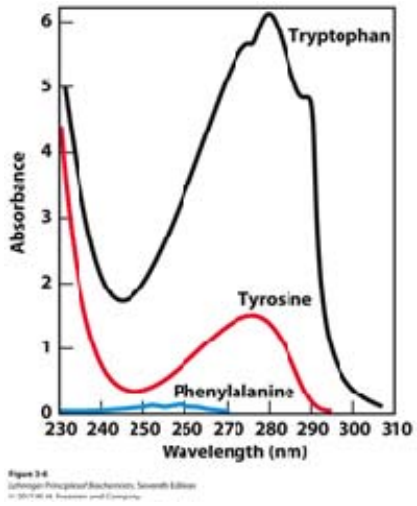
- **Nonpolar Amino Acids:**

  - They contain hydrocarbon chains, so their ability to interact with water is very limited. They are, therefore, hydrophobic and avoid interactions with water. In this group are **Gly, Ala, Pro, Val, Leu, Ile, and Met**.
  - They play an important role in maintaining the three-dimensional structure of the protein based on the hydrophobic interactions they generate.
  - **Glycine** is the only one without stereoisomerism.



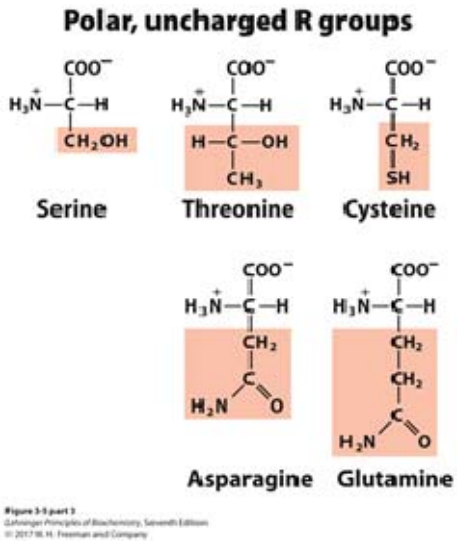
- **Aromatic Amino Acids: Phe, Tyr, and Trp** are relatively nonpolar.
  - **Tyr**, due to the presence of the -OH group, has polarity and is likely to be reversibly phosphorylated in a process mediated by kinases and phosphatases. This chemical modification is a mechanism for regulating the activity of the proteins in which it occurs.
  - The presence of aromatic groups gives these AAs the property of absorbing ultraviolet light. This particularity is especially notable in the case of **Tyr and Trp** (with an absorption peak at 270 nm) and less evident in the case of **Phe** (with a peak at 260 nm).





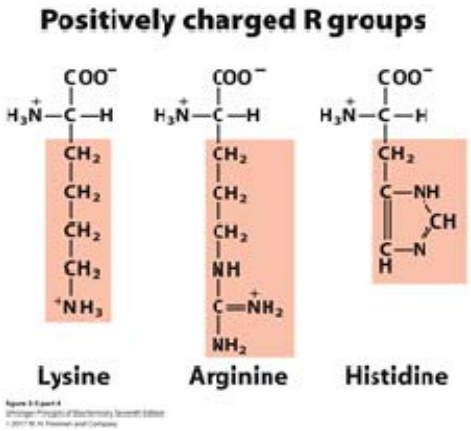
- **Uncharged Polar Amino Acids:** They are hydrophilic, with a great capacity to establish interactions with water and with other side chains through hydrogen bonds. **Ser, Thr, Cys, Asn, and Gln** belong to this group.

- **Cys** has a special role in this group. The sulfhydryl group ($-\text{SH}$) is capable of oxidizing and forming a covalent bond (disulfide bridge) that joins two Cys residues. These types of bonds are relevant in joining different polypeptide chains or in stabilizing the conformation of a single polypeptide chain.

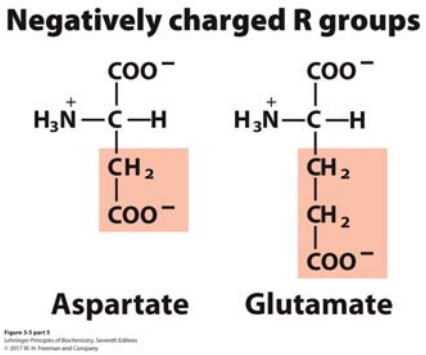


- As previously mentioned regarding the reversible phosphorylation of Tyr, the same happens with **Ser and Thr**. The presence of -OH groups in their side chains allows their phosphorylation, which impacts protein activity.

- **Positively Charged Amino Acids (Basic):** They are also polar. **Lys, Arg, and His** belong to this group.
  - The presence of positive charges allows them to establish ionic interactions with other AAs or molecules that have negative charges.



- **Negatively Charged Amino Acids (Acidic):** They offer the possibility of establishing ionic interactions. **Asp and Glu** belong to this group.



Aa are acids and bases

- Aa behave like acids and bases
- The α-amino and α-carboxyl groups, as well as the ionizable R groups of some Aa are weak acids and bases
- These groups, depending on the pH of the medium, will be protonated or deprotonated modifiying their net charge.
- The simplest case involves Aa with non-ionizable side chains, such as Gly
- It has two groups, –COOH and $–^{+}NH_3$, which can release protons.
- Therefore, depending on the pH, Gly can have a +1 charge (at a very low pH), 0 (at a pH close to neutrality) and -1 (at a basic pH).
- In the titration curve, we can observe pH variations depending on the concentration of base (OH^-) added. Three intermediate points are distinguished during the ionization of each of the ionizable groups:

First intermediate point: Maximum buffering capacity, corresponding to the pK of the –COOH group. At this point, there are equal concentrations of the –COOH and $–COO^-$ species.

Isoelectric Point (pI) The slope of the curve reaches its maximum because most of the AA is found as $^{-}OOC\text{-}CH_2\text{-}^{+}NH_3$. This species has a net charge of 0 and is called a **zwitterion**. The pH value at which this point is reached is called the **isoelectric point**

Last intermediate point: Corresponds to the ionization of the amino group ($-^{+}NH_3$)

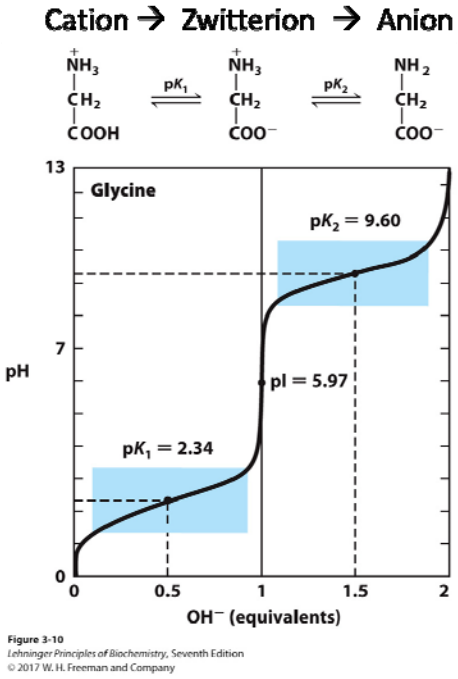
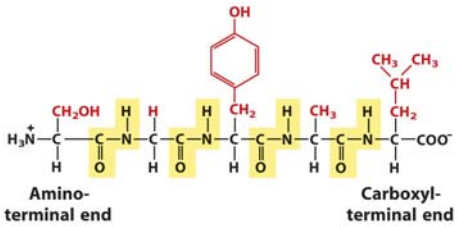


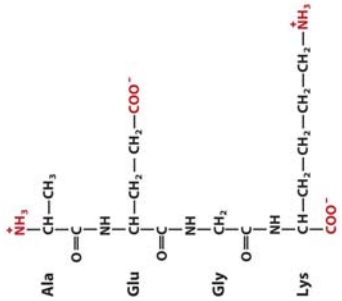
Figure 3-10
Lehninger Principles of Biochemistry, Seventh Edition
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Here we again find a phase where equilibrium occurs between the $-^{+}NH_3$ - NH_2 species.

Peptides and Proteins



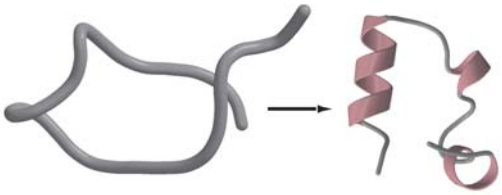
- By consensus, we use the term **peptides** to refer to small polypeptide chains or fragments derived from the proteolysis of larger proteins.
- Peptides originate through the polymerization of AAs.
- The bond that joins two AAs together is called a **peptide bond**.
- The bond is produced by a condensation reaction between the carboxyl group of one AA and the amino group of another.
- Under normal conditions, the spontaneous formation of this bond is highly thermodynamically unfavorable.
- Cells solve this problem by activating the carboxyl group through the formation of an ester bond between the carboxyl group and an OH at the 3' end of the tRNA.
- The free carboxyl group of the growing peptide then reacts with another amino group, and the chain continues to extend.
- Once a small polymer of AAs is formed, we can clearly distinguish two distinct termini:
    - **C-terminus (C-ter):** The end with the free carboxyl group.
    - **N-terminus (N-ter):** The end with the free amino group.
    - Each of the incorporated amino acids is called a **residue**.



- A characteristic of all proteins is their net charge, which depends on their AA composition. The pH at which a protein lacks a net charge is a characteristic parameter called the **isoelectric point**.
- Size is a highly variable parameter:
    - There are active peptides with only 9 AAs (e.g., oxytocin).
    - Other proteins have several hundred or even thousands of AAs.
- Structure and composition variations:
    - Some proteins consist of a single polypeptide chain (monomeric), while others have multiple chains (multimeric).
    - Amino acid composition varies significantly between different proteins.

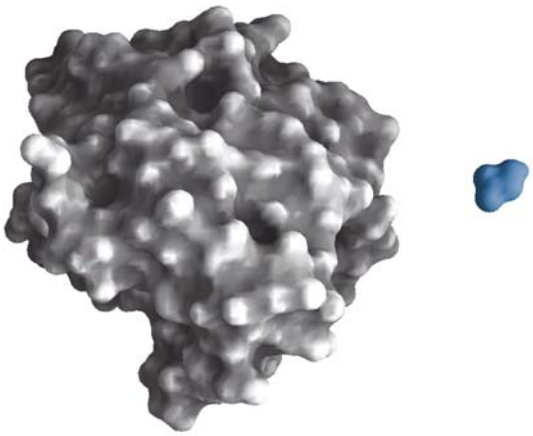
- Conjugated proteins may have non-protein chemical moieties attached that are essential for their biological activity.

Tridimensional structure of proteins



The single bonds making up the backbone of a protein allow it to adopt an almost unlimited number of conformations.

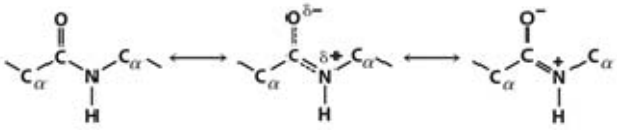
- Each protein has a unique conformation in which it exhibits activity.
- Conformation is a direct consequence of the amino acid sequence forming the protein.
- **Conformation:** The spatial arrangement of the atoms in a protein.
- Proteins can change their conformation without breaking covalent bonds as a result of chemical modifications or changes in physicochemical conditions. These conformational changes often impact protein activity.
- The active conformation of a protein is called its **native state**.
- The stabilization of the native conformation is mainly driven by:
  - Hydrogen bonds
  - Ionic interactions
  - Hydrophobic interactions
  - Disulfide bonds



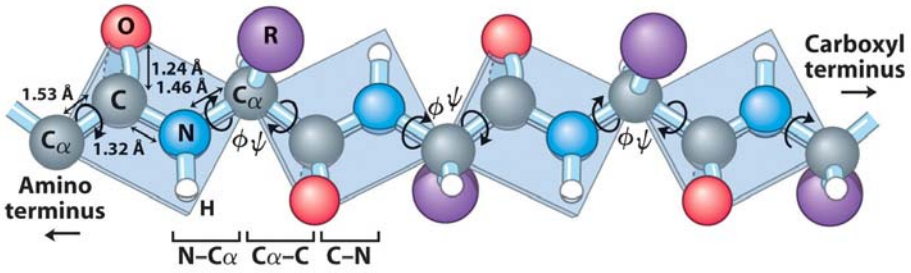
- Although disulfide bridges are individually much stronger than non-covalent interactions, the sheer number of weak interactions present in a protein makes them collectively more significant in maintaining three-dimensional structure.

Characteristics of the peptide bond

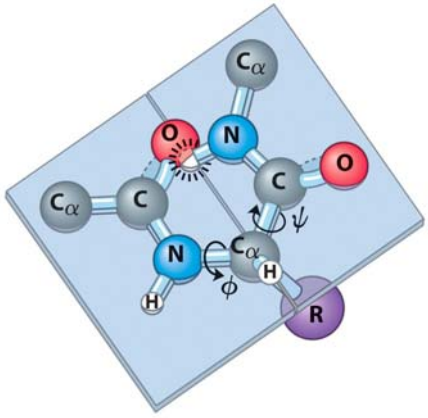
- The first structural studies of proteins were carried out by **Linus Pauling and Robert Corey in 1930**.
- They studied the fundamental characteristics of the peptide bond.
- The Cα atoms of two contiguous amino acids are separated by three covalent bonds: Cα-C-N-Cα..



- X-ray diffraction showed that the C-N peptide bond is shorter than a standard amine single bond and that both participating atoms lie in the same plane.
- Carbonyl oxygen and amide nitrogen share electron density, giving the C-N bond partial double-bond character.
- The six atoms of the peptide group lie in the same plane, with the carbonyl oxygen and amide hydrogen in a *trans* configuration.
- Due to these characteristics, **rotation around the C}-N peptide bond is restricted/not possible**.
- Rotation **is possible** around the N- Cα and Cα-C bonds:
- The backbone of a protein is thus a succession of rigid planes linked at pivot points Cα.

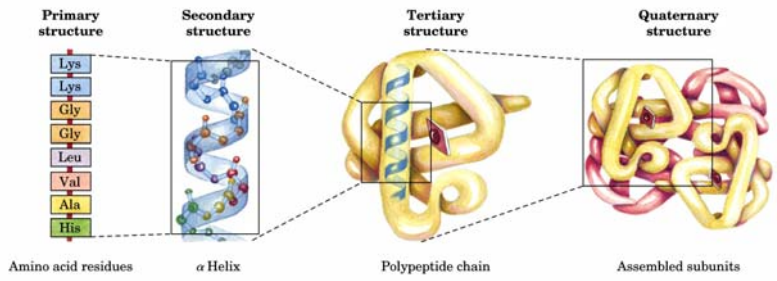


- The dihedral angle of the N- Cα bond is designated φ **(phi)**, and that of the Cα-C bond, ψ **(psi)**.
- Theoretically, these single bonds could adopt any angle between -180º and +180º. However, many values are forbidden due to steric hindrance between backbone atoms and side chains.



- From its primary amino acid sequence—encoded by genetic information—to its fully

folded native state, a protein adopts increasingly complex structural levels (up to four levels).

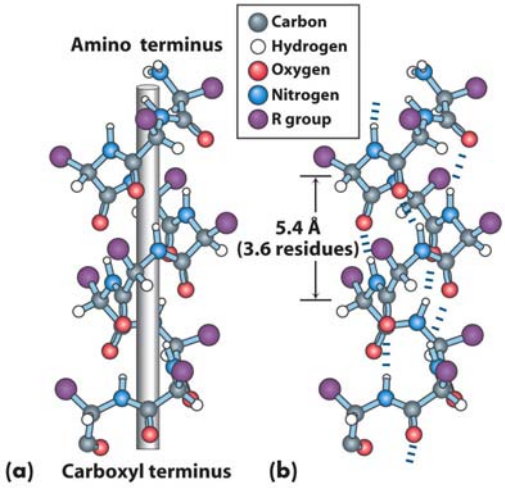


The Secondary structure of Proteins

Secondary structure refers to the **local conformation of segments of the polypeptide chain**, usually resulting from regular folding patterns. The most common and stable types are **α helices and β conformations** (predicted by Pauling and Corey in 1951).

The α helix

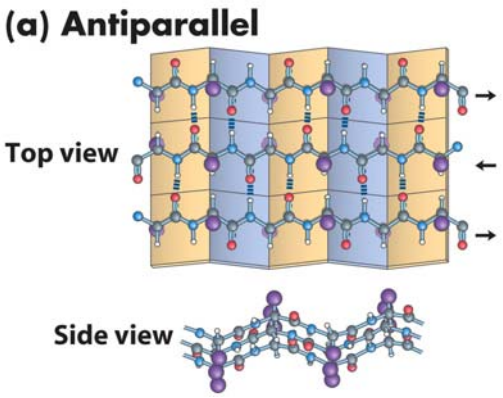
- One of the most abundant secondary structure elements.
- In structural proteins (e.g., alpha-keratins in hair and nails), it is almost the sole secondary structure element present.
- In globular proteins (e.g., enzymes, receptors), it coexists with other elements like beta-conformations.
- Consists of a polypeptide backbone tightly wound around a central longitudinal axis, with R groups protruding outward.



- Highly stable because all carbonyl oxygens and amide hydrogens form regular, intrachain hydrogen bonds.

β-Conformation and β-Sheets

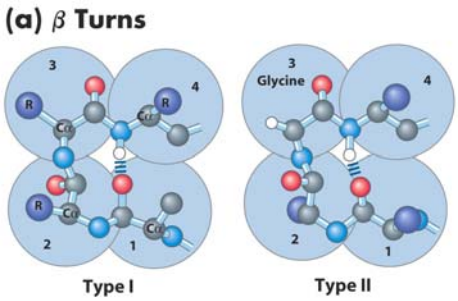
- Polypeptide chains are extended in a zigzag conformation.
- Adjacent extended segments line up side-by-side to form hydrogen bonds, creating a **β-sheet**.
- Side chains (R groups) extend perpendicularly from the zigzag plane in alternating directions.



- Depending on chain orientation, β-sheets are **parallel** or **antiparallel**.

- Most frequent AAs in β-sheets are **Gly and Ala** due to their small side chains, allowing tight packing of adjacent sheets.

β-Turns



- Compact folding of polypeptide chains requires frequent, sharp changes in direction.
- In antiparallel β-conformations, connections between adjacent segments are formed by **β-turns**.
- Composed of **four amino acid residues**, stabilized by a hydrogen bond between the carbonyl oxygen of the 1st residue and the amide hydrogen of the 4th residue.
- **Gly** (small size) and **Pro** (imino ring facilitating tight turns) are frequently present.

- Four Aa are involved in this structure, and sustained by a hydrogen bond between the carbonyl oxygen of the first Aa and the amino hydrogen of the fourth
- The Gly and the Pro are frequent, the first because of its small size and the second because it facilitates the turning of the chain because of the imino group.

Tertiary and quaternary structures

Tertiary Structure

- Refers to the three-dimensional arrangement of all atoms in a single polypeptide chain.



- Polypeptide chains fold compactly, bringing distantly spaced primary sequence residues into close spatial proximity to form active interactions.

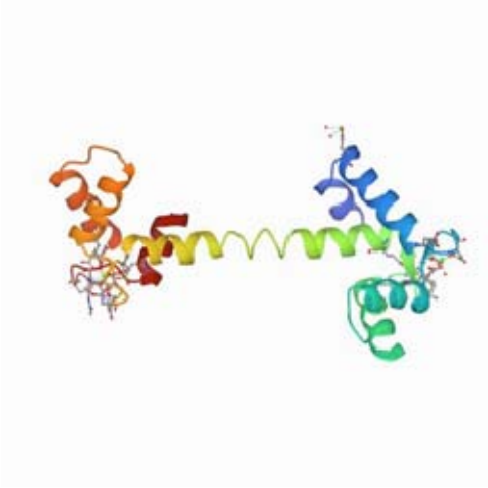
- Combines various secondary structural elements α-helices, β-sheets, turns, and loops).
- Stabilized by intramolecular forces: hydrogen bonds, electrostatic interactions, hydrophobic effects, and disulfide bridges.
- Folding creates functional sites: ligand-binding pockets, interaction domains, subcellular localization signals, etc. These frequently occur within distinct **structural domains**.

TABLE 4–2 Approximate Amounts of a Helix and b Conformation in Some Single-Chain Proteins

	Residues (%)*	
Protein (total residues)	*α Helix*	*β Conformation*
Chymotrypsin (247)	14	45
Ribonuclease (124)	26	35
Carboxypeptidase (307)	38	17
Cytochrome *c* (104)	39	0
Lysozyme (129)	40	12
Myoglobin (153)	78	0

Source: Data from Cantor, C.R. & Schimmel, P.R. (1980) *Biophysical Chemistry*, Part I: *The Conformation of Biological Macromolecules*, p. 100, W. H. Freeman and Company, New York.

*Portions of the polypeptide chains that are not accounted for by α helix or β conformation consist of bends and irregularly coiled or extended stretches. Segments of α helix and β conformation sometimes deviate slightly from their normal dimensions and geometry.

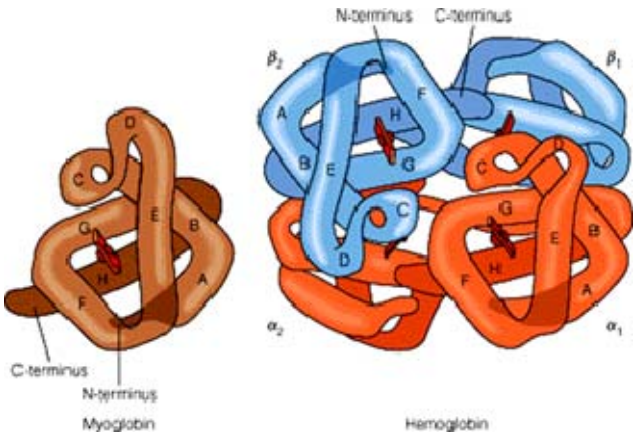


Quaternary Structure

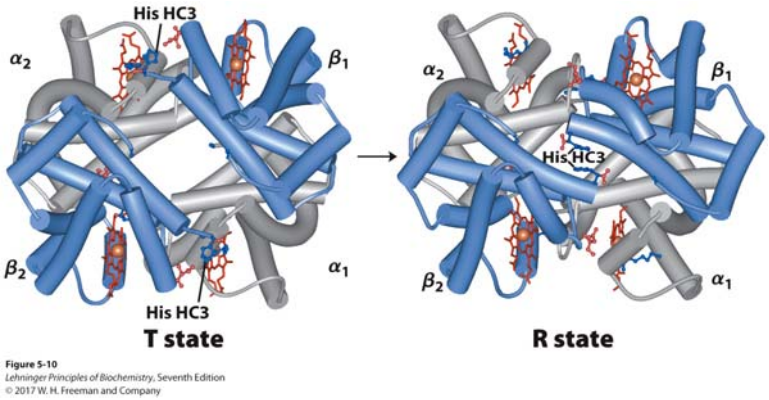
- Present when a protein consists of two or more polypeptide chains (identical or distinct) interacting with each other. Each chain is termed a **monomer** or **subunit**.
- Assembly into quaternary structures can generate novel functional properties not present in isolated monomers (e.g., cooperativity).

Myoglobin vs. Hemoglobin:

- **Myoglobin:** Monomeric oxygen-storage protein in muscle (153 AAs), single chain with a heme group, predominantly α-helical with a hydrophobic core and a proximal His bound to Fe^{2+}).



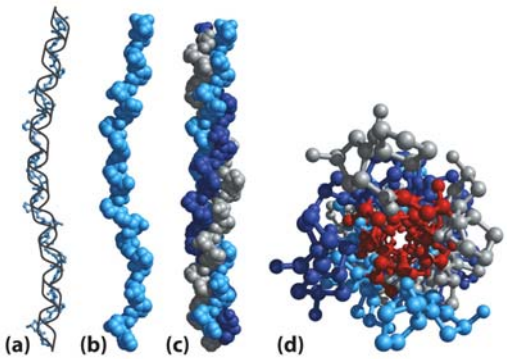
- **Hemoglobin:** Tetrameric transport protein α2/β2) containing four heme groups. Oxygen binding induces conformational transitions between two distinct quaternary states (T and R states with differing O_2 affinities).

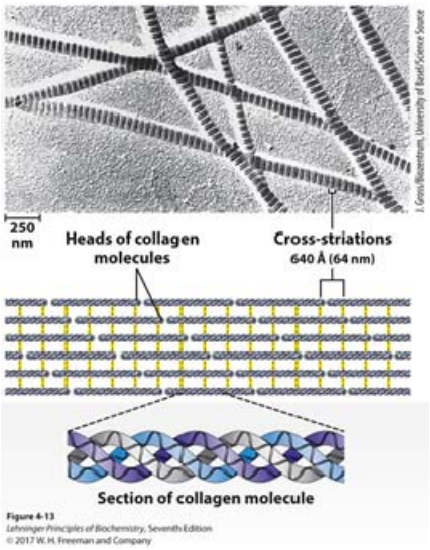


Classification of Proteins by Structure

- **Fibrous Proteins:** Polypeptide chains arranged in elongated strands or sheets, dominated by a single repeating secondary structure. They provide strength, elasticity, and are insoluble due to high hydrophobic content.
- **Collagen:** Found in connective tissue (tendons, cartilage, bone, cornea). Forms a unique left-handed helix (3 AAs per turn) rich in Gly, Pro, Hyp (hydroxyproline), and Hyl (hydroxylysine). Three chains wrap into a right-handed superhelix (**tropocollagen**). Hydroxylation of Pro/Lys requires **Vitamin C** as an essential cofactor; deficiency leads to scurvy due to compromised triple-helix stability.

  Note: In Ehlers-Danlos syndrome, substitution of critical Gly residues by larger





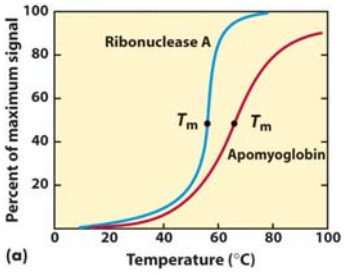
AAs (Cys, Ser) disrupts packing and leads to joint hypermobility.

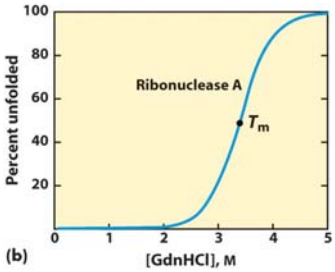
- **Globular Proteins:** Polypeptide chains compactly folded into spherical or globular shapes, combining multiple secondary structure types. They perform diverse dynamic functions (enzymes, transport, motor proteins, immunoglobulins).

Protein Denaturation

The native conformation of a protein is maintained within narrow physiological ranges of temperature, salt concentration, and pH.

- Extreme conditions cause loss of three-dimensional structure and loss of biological activity (**denaturation**).
- **Thermal Denaturation:** Heat denaturation exhibits cooperative kinetics (slow initial phase followed by rapid unfolding), indicating that disruption of one region destabilizes adjacent structural elements.
- **Chemical Denaturation:** Chaotropic agents like **guanidinium chloride** or **urea** disrupt hydrophobic interactions and hydrogen bonding networks, leading to unfolding.





Renaturation (Anfinsen's Experiment)

- Treatment of ribonuclease with urea and a reducing agent (β-mercaptoethanol) results in complete unfolding and loss of activity.
- Removal of urea and reducing agent allows spontaneous refolding into the active native structure with correct disulfide bond pairing.
- **Conclusion:** All information required for a protein to fold into its native three-dimensional structure is contained within its **primary amino acid sequence**.

